UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	DATABASE DESIGN AND DEVELOPMENT
COURSE CODE	:	ICT450
EXAMINATION	:	2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the ICT450 mock paper (2026).

2. Five questions x 20 marks = 100 marks.
Questions and answers are for local study use only.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

QUESTION 1

Database Concept

a) A database is an organised collection of related data. A DBMS is software that creates, stores, manages and controls access to a database. A database system is the database, DBMS, application programs, people, procedures and hardware working together.

Marking points: database definition (2); DBMS definition (2); database system definition (2)

(6 marks)

b) The limitations include data redundancy, data inconsistency, difficult data sharing, weak concurrency control, limited security, program-data dependence, and difficult backup and recovery.

Marking points: four limitations x 2 marks each

(8 marks)

c i) Candidate keys: Customer No, Customer IC and e-mail (any TWO). Primary key: Customer No, because it is unique for every row and does not change.

Marking points: two candidate keys (2); primary key with reason (1)

(3 marks)

c ii) A spreadsheet stores unrelated copies of data, so the same customer can be repeated and updated inconsistently. It lacks integrity constraints, concurrency control, access control and recovery facilities that a DBMS provides.

Marking points: redundancy and inconsistency (1); missing integrity constraints (1); missing concurrency, security or recovery (1)

(3 marks)

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QUESTION 2

Relational Data Model

a) STUDENT: primary key Student_ID; candidate keys Student_IC, Student_ID and EMAIL.
COURSE: primary key COURSE_ID.
Marking points: STUDENT primary key (1); two candidate keys (2); COURSE primary key (1)

(4 marks)

b) Entity integrity: a primary-key column cannot contain a null or duplicate value. Referential integrity: a foreign-key value must match an existing primary-key value or be null. Example: every COURSE_ID in STUDENT must exist in COURSE, and STUDENT_ID must be unique and not null.
Marking points: entity integrity definition (2); referential integrity definition (2); example from the tables (2)

(6 marks)

c) Tuple = a row; attribute = a column; degree = number of attributes; cardinality = number of tuples.
Marking points: one mark each

(4 marks)

d) If COURSE_NAME is repeated for every student, changing a course name in one row but not the others causes an update anomaly. Deleting the last student for a course loses the course data (delete anomaly), and adding a new course before any student enrols is difficult (insert anomaly).
Marking points: two anomalies explained x 3 marks

(6 marks)

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QUESTION 3

ERD

a) Entities: VENDOR with VEND_CODE, VEND_CONTACT, VEND_AREACODE and VEND_PHONE; PRODUCT with PROD_CODE, PROD_DESCRIPT, PROD_PRICE, PROD_ON_HAND and VEND_CODE. Relationship: VENDOR supplies PRODUCT. Cardinality: one VENDOR supplies many PRODUCTs (1:M).

Marking points: entities (2); attributes (2); relationship (2); cardinality (2)

(8 marks)

b) PRODUCT.VEND_CODE is a foreign key that must match an existing VENDOR.VEND_CODE or be null. Inserting a PRODUCT with a VEND_CODE that does not exist in VENDOR violates referential integrity; deleting a VENDOR that still has PRODUCTs also violates it unless the products are deleted or reassigned.

Marking points: foreign-key constraint (3); violation example (3)

(6 marks)

c) VENDOR(VEND_CODE PK, VEND_CONTACT, VEND_AREACODE, VEND_PHONE); PRODUCT(PROD_CODE PK, PROD_DESCRIPT, PROD_PRICE, PROD_ON_HAND, VEND_CODE FK REFERENCES VENDOR(VEND_CODE)).

Marking points: VENDOR relation and primary key (3); PRODUCT relation, primary key and foreign key (3)

(6 marks)

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QUESTION 4

Normalization

a) Relation: COURSE_RECORD(CourseCode, CourseName, TutorID, TutorName, StudentID, StudentName, DOB, Gender, LastAttendance). Candidate key: (CourseCode, StudentID). FDs: CourseCode -> CourseName, TutorID, TutorName; StudentID -> StudentName, DOB, Gender; (CourseCode, StudentID) -> LastAttendance.

Marking points: candidate key (2); functional dependencies (2)

(4 marks)

b) 1NF: all attributes are atomic. 2NF: 1NF and every non-key attribute is fully functionally dependent on the whole candidate key. 3NF: 2NF and no non-key attribute is transitively dependent on the key.

Marking points: 1NF (2); 2NF (2); 3NF (2)

(6 marks)

c) Partial: CourseCode -> CourseName, TutorID, TutorName depends on part of the key; StudentID -> StudentName, DOB, Gender depends on part of the key. Transitive: CourseCode -> TutorID -> TutorName.

Marking points: partial dependencies (2); transitive dependency (2)

(4 marks)

d) COURSE(CourseCode PK, CourseName, TutorID FK); TUTOR(TutorID PK, TutorName); STUDENT(StudentID PK, StudentName, DOB, Gender); ATTENDANCE(CourseCode PK, StudentID PK, LastAttendance, FKs to COURSE and STUDENT).

Marking points: four relations (4); primary and foreign keys (2)

(6 marks)

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QUESTION 5

Database Design

a) Conceptual design: identify entities, attributes and relationships and build a conceptual model. Logical design: map the conceptual model to a relational schema, choose keys and normalize. Physical design: choose storage structures, indexes, file organisation and access paths. Implementation: create the database objects and load the data.

Marking points: conceptual design (2); logical design (2); physical design (2); implementation (2)

(8 marks)

b) SDLC covers the whole information-system life cycle; DBLC focuses on the database component within that system and runs through its own design, implementation, testing, operation and maintenance phases.

Marking points: SDLC scope (2); DBLC scope (2)

(4 marks)

c) Top-down starts from the overall data model and breaks it into components. Bottom-up starts from individual data elements and combines them into a model.

Marking points: top-down (2); bottom-up (2)

(4 marks)

d) Centralized design is controlled by one design team or location. Decentralized design distributes design work across teams or locations and requires reconciliation.

Marking points: centralized (2); decentralized (2)

(4 marks)

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